

AD165D Datasheet

Zhuhai Jieli Technology Co.,LTD

Version: 1.0

Date: 2024.04.01

AD165D Features

CPU

- 32bit DSP
- Maximum speed 160MHz
- Interrupts with 8 priority level

Memory

- OTP
- Optional built-in flash memory

Clocks

- On-chip 16 MHz clock
- On-chip 200KHz lower-temperature-drift clock
- 32.768 KHz crystal oscillator

DSP Audio Processing

- Support MP2, MP3, WMA, WAV decoding
- Multi-band DRC limiter
- Multi-band EQ configuration for voice Effects

Audio Codec

- Two channels 16-bit DAC, single-ended with SNR ≥ 93 dB, differential with SNR ≥ 100 dB
- One channel 24-bit ADC, SNR ≥ 85 dB
- Audio DAC Sampling rates of 8KHz/11.025KHz/16KHz/22.05KHz/24KHz/32KHz/44.1KHz/48KHz/64KHz/88.2KHz/96KHz are supported
- Audio ADC Sampling rates of 8KHz/11.025KHz/16KHz/22.05KHz/24KHz/32KHz/44.1KHz/48KHz are supported
- Audio DAC support single-ended and differential cap-less mode
- Support analog audio input
- Support for driving 16 or 32 ohm speaker

Peripherals

- One full speed USB OTG controller
- One SD host controller for MMC/SD
- Three multi-function 32-bit timers, support capture and PWM mode
- UART0 controller
- The UART1 supports DMA and flow control
- One IIC Master controller
- Two SPI Master / Slaver controller with DMA
- One QDEC interface
- 12-channel 10-bit general purpose ADC
- 4-channel Advance PWM controller
- LCD controller
- 18 Individually programmable and multiplexed GPIO pins
- Digital peripheral crossbar
- Up to 12 external interrupt / wake-up source (low power available, can be multiplexed to any I/O)

PMU

- RTC Alarm Wakeup
- Less than 2uA soft off current
- VBAT range : 2.2V to 5.0V
- IOVDD range : 2.1V to 3.6V

Packages

- QSOP24

Temperature

- Operating temperature: -40°C to +85°C
- Storage temperature: -65°C to +150°C

Applications

- Audio player
- Microcontrollers

1 Block Diagram

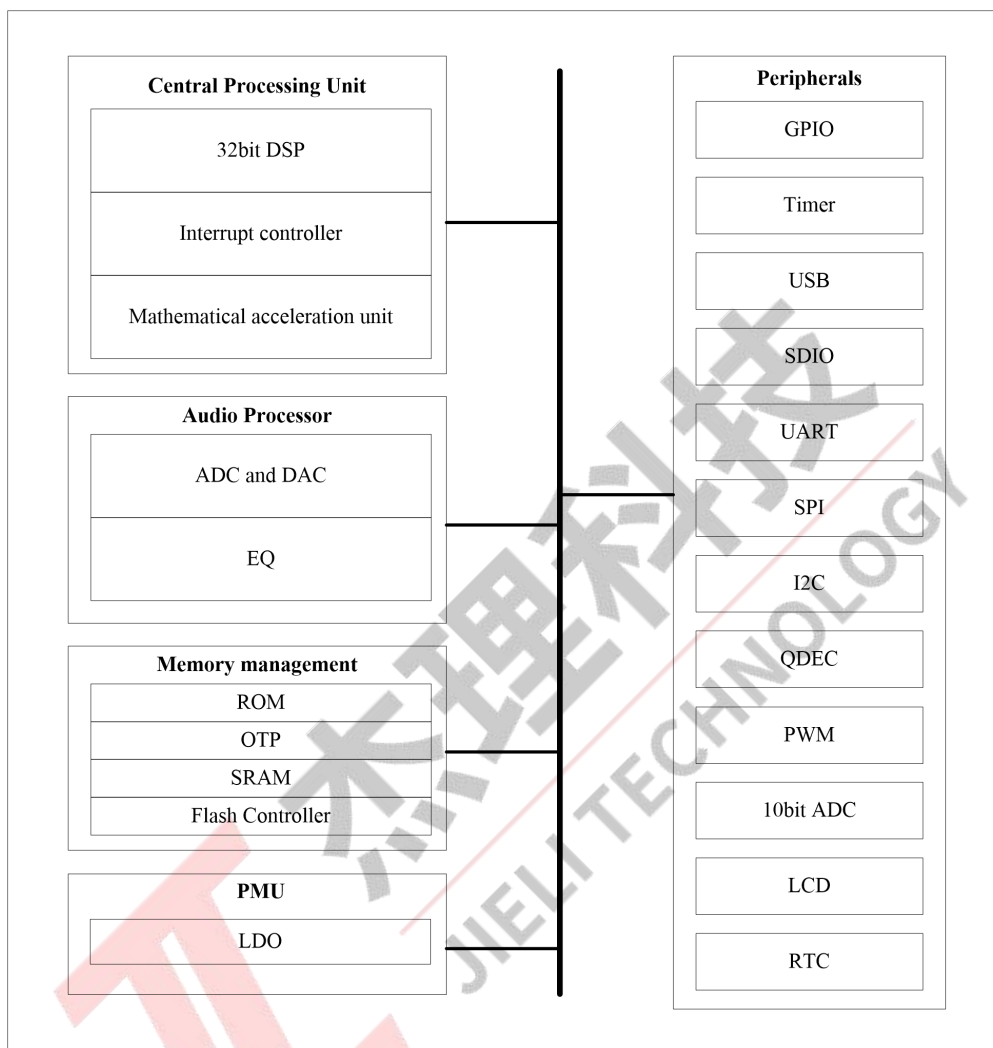


Figure 1-1 AD165D Block Diagram

2 Pin Definition

2.1 Pin Assignment

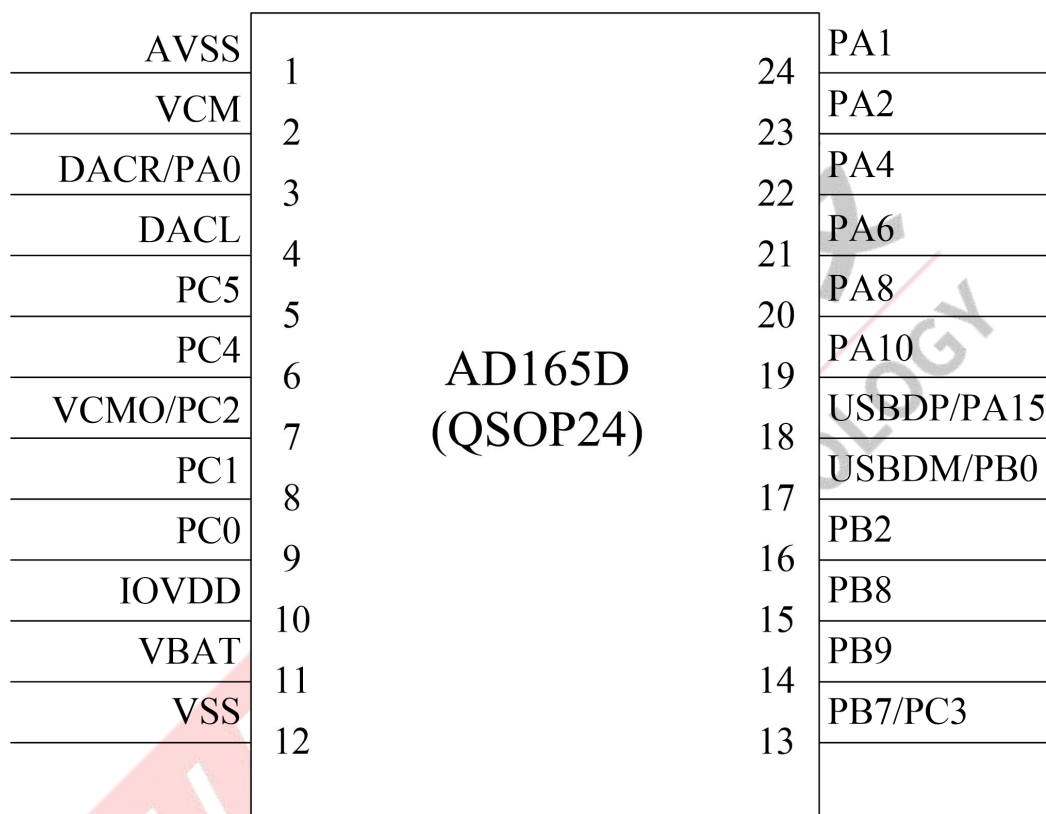


Figure 2-1 AD165D Package Diagram

2.2 Pin Description

Table 2-1 AD165D Pin Description

PIN NO.	Name	Type	Function	Other Function
1	AVSS	G		Audio ground;
2	VCM	P		Audio analog reference bias;
3	PA0	I/O	GPIO	MICLDO:Microphone linear voltage regulator output; ADC0:ADC Input Channel 0; LCD SEG0;
	DACR	AO		Right channel audio output;
4	DACL	AO		Left channel audio output;
5	PC5	I/O	GPIO	AINR:Right channel analog audio input; LCD COM0(A);
6	PC4	I/O	GPIO	AINL:Left channel analog audio input; SPI0CS(B):SPI0 Chip Select(B); LCD COM1(A);
7	VCMO	AO		negative of earphone;
	PC2	I/O	GPIO	LCD COM3(A);
8	PC1	I/O	GPIO	SPI0CLK(B):SPI0 Clk(B); ADC12:ADC Input Channel 12; LCD COM4(A);
9	PC0	I/O	GPIO	SPI0DO(B):SPI0 Data Out(B); ADC11:ADC Input Channel 11; LCD SEG26; LCD COM5(A);
10	IOVDD	PO	Power supply for GPIO	Built-in linear voltage regulator output;
11	VBAT	P		Battery interface;
12	VSS	G		System ground;
13	PC3	I/O	GPIO	SPI0DI(B):SPI0 Data In(B); SDPG:SD card Power Gate; ADC13:ADC Input Channel 13; LCD COM2(A);
	PB7	I/O	GPIO	ADC9:ADC Input Channel 9; LCD SEG23;
14	PB9	I/O	GPIO	ROSCI_32K:32.768KHz crystal oscillator input; LCD SEG25;
15	PB8	I/O	GPIO	ROSCO_32K:32.768KHz crystal oscillator output; ADC10:ADC Input Channel 10; LCD SEG24;
16	PB2	I/O	GPIO	ADC6:ADC Input Channel 6; LCD SEG18;

17	PB0	I/O	GPIO	ADC5:ADC Input Channel 5; LCD SEG16;
	UDBDM	I/O	USB Negative Data (pull down)	ADC15:ADC Input Channel 15;
18	USBDP	I/O	USB Positive Data (pull down)	ADC14:ADC Input Channel 14;
	PA15	I/O	GPIO	LCD SEG15;
19	PA10	I/O	GPIO	ADC4:ADC Input Channel 4; LCD SEG10;
20	PA8	I/O	GPIO (pull up)	Long press reset; ADC3:ADC Input Channel 3; LCD SEG8;
21	PA6	I/O	GPIO	ADC2:ADC Input Channel 2; LCD SEG6;
22	PA4	I/O	GPIO	ADC1:ADC Input Channel 1; LCD SEG4;
23	PA2	I/O	GPIO	MICIN1:MIC1 Input Channe; LCD SEG2;
24	PA1	I/O	GPIO	MICIN0:MIC0 Input Channe; TMR0:Timer0 Clock Input; LCD SEG1;

Pin Type	Description	Pin Type	Description
P	Power	I/O	Input or Output
PI	Power Input	I	Input
PO	Power Output	O	Output
AO	Analog Output	G	Ground

CROSSBAR						
SPI1	SDC	IIC	UART0	UART1	Q-decoder	MCPWM
SPI1_CLK	SD_CLK	IIC_CLK	UART0_TX	UART1_TX	Q-decoder A	PWMCH0L
SPI1_DI	SD_CMD	IIC_DAT	UART0_RX	UART1_RX	Q-decoder B	PWMCH0H
SPI1_DO	SD_DAT			UART1_CTS		PWMCH1L
				UART1_RTS		PWMCH1H

Input Channel x10		Output Channel x8	
CAP0	Timer0	PWM0	CLK_OUT0
CAP1	Timer1	PWM1	CLK_OUT1
CAP2	Timer2	PWM2	CLK_OUT2
		LEDC	CLK_OUT3

3 Electrical Characteristics

3.1 Absolute Maximum Ratings

Table 3-1

Symbol	Parameter	Min	Max	Unit
T _{opt}	Operating temperature	-40	+85	°C
T _{stg}	Storage temperature	-65	+150	°C
V _{BAT}	Supply Voltage	-0.3	5.0	V
V _{IOVDD}	Voltage applied at IOVDD	-0.3	3.6	V
V _{GPIO}	Voltage applied to GPIO	-0.3	IOVDD+0.3	V

Note : The chip can be damaged by any stress in excess of the absolute maximum ratings listed below

3.2 PMU Characteristics

Table 3-2

Symbol	Parameter	Min	Typ	Max	Unit	Test Conditions
V _{BAT}	Voltage Input	2.2	3.7	5.0	V	—
IOVDD	Voltage output	2.1	3.0	3.6	V	V _{BAT} = 4.2V, 10mA loading
	Loading current	—	—	100	mA	IOVDD=3.3V@V _{BAT} = 3.6V
V _{LVD}	Voltage input	2.1	2.8	2.8	V	Low-Voltage Detection of IOVDD

3.3 IO Input/Output Electrical Logical Characteristics

Table 3-3

GPIO input characteristics						
Symbol	Parameter	Min	Typ	Max	Unit	Test Conditions
V _{IL}	Low-Level Input Voltage	-0.3	—	0.3* IOVDD	V	IOVDD = 3.0V
V _{IH}	High-Level Input Voltage	0.7* IOVDD	—	IOVDD+0.3	V	IOVDD = 3.0V
GPIO output characteristics						
Symbol	Parameter	Min	Typ	Max	Unit	Test Conditions
V _{OL}	Low-Level Output Voltage	—	—	0.1* IOVDD	V	IOVDD = 3.0V
V _{OH}	High-Level Output Voltage	0.9* IOVDD	—	—	V	IOVDD = 3.0V

3.4 Internal Resistor Characteristics

Table 3-4

Port	Drive Current	Internal Pull-Up Resistor	Internal Pull-Down Resistor	Comment
PA0~PA2 PA4,PA6,PA8 PA10,PA15 PB0,PB2 PB7~PB9 PC0~PC5	2mA(HD1,HD0==0,0) 5.6mA(HD1,HD0==0,1) 18mA(HD1,HD0==1,0) 30mA(HD1,HD0==1,1)	10K	10K	1. PA8 default pull up 2. USBDM,USBDP default pull down 3. Internal pull-up/pull-down resistance accuracy $\pm 20\%$
USBDP	27mA	1.5K	15K	
USBDM		180K	15K	

3.5 Audio DAC Characteristics

Table 3-5

Parameter	MODE	Min	Typ	Max	Unit	Test Conditions
Frequency Response		20	—	20K	Hz	1KHz/0dB 10k ohm loading With A-Weighted Filter IOVDD>2.7V
Output Swing	Diff (R to L)	—	1.5	—	Vrms	
	Single-ended	—	750	—	mVrms	
THD+N	Diff (R to L)	—	-80	—	dB	
	Single-ended	—	-80	—	dB	
S/N	Diff (R to L)	—	100	—	dB	
	Single-ended	—	97	—	dB	
Dynamic Range	Diff (R to L)	—	100	—	dB	1KHz/-60dB 10k ohm loading With A-Weighted Filter IOVDD>2.7V
	Single-ended	—	93	—	dB	
Noise Floor	Diff (R to L)	—	13	—	uVrms	With A-Weighted Filter IOVDD>2.7V
	Single-ended	—	11	—	uVrms	
Crosstalk	Single-ended	—	-93	—	dB	10KHz/0dB 10k ohm loading IOVDD>2.7V
	(R and L) to VCMO	—	-60	—	dB	10KHz/0dB 32 ohm loading IOVDD>2.7V
	(R and L) to VCMO	—	-57	—	dB	10KHz/0dB 16 ohm loading IOVDD>2.7V

3.6 Audio ADC Characteristics

Table 3-6

Parameter	MODE	Min	Typ	Max	Unit	Test Conditions
Dynamic Range	Differential	—	88	—	dB	Fsample=44.1KHz,Gain=4dB Fin=1KHz @1Vpp IOVDD>2.7V,With A-Weighted Filter
		—	83	—	dB	Fsample=44.1KHz,Gain=20dB Fin=1KHz @160mVpp IOVDD>2.7V,With A-Weighted Filter
	Single-ended	—	88	—	dB	Fsample=44.1KHz,Gain=-2dB Fin=1KHz @1Vpp IOVDD>2.7V,With A-Weighted Filter
		—	76	—	dB	Fsample=44.1KHz,Gain=14dB Fin=1KHz @160mVpp IOVDD>2.7V,With A-Weighted Filter
S/N	Differential	—	88	—	dB	Fsample=44.1KHz,Gain=4dB Fin=1KHz @1Vpp IOVDD>2.7V,With A-Weighted Filter
		—	80	—	dB	Fsample=44.1KHz,Gain=20dB Fin=1KHz @160mVpp IOVDD>2.7V,With A-Weighted Filter
	Single-ended	—	88	—	dB	Fsample=44.1KHz,Gain=-2dB Fin=1KHz @1Vpp IOVDD>2.7V,With A-Weighted Filter
		—	76	—	dB	Fsample=44.1KHz,Gain=14dB Fin=1KHz @160mVpp IOVDD>2.7V,With A-Weighted Filter
THD+N	Differential	—	-80	—	dB	Fsample=44.1KHz,Gain=4dB Fin=1KHz @1Vpp IOVDD>2.7V,With A-Weighted Filter
		—	-78	—	dB	Fsample=44.1KHz,Gain=20dB Fin=1KHz @160mVpp IOVDD>2.7V,With A-Weighted Filter
	Single-ended	—	-78	—	dB	Fsample=44.1KHz,Gain=-2dB Fin=1KHz @1Vpp IOVDD>2.7V,With A-Weighted Filter
		—	-70	—	dB	Fsample=44.1KHz,Gain=14dB Fin=1KHz @160mVpp IOVDD>2.7V,With A-Weighted Filter

4 Package Information

4.1 QSOP24

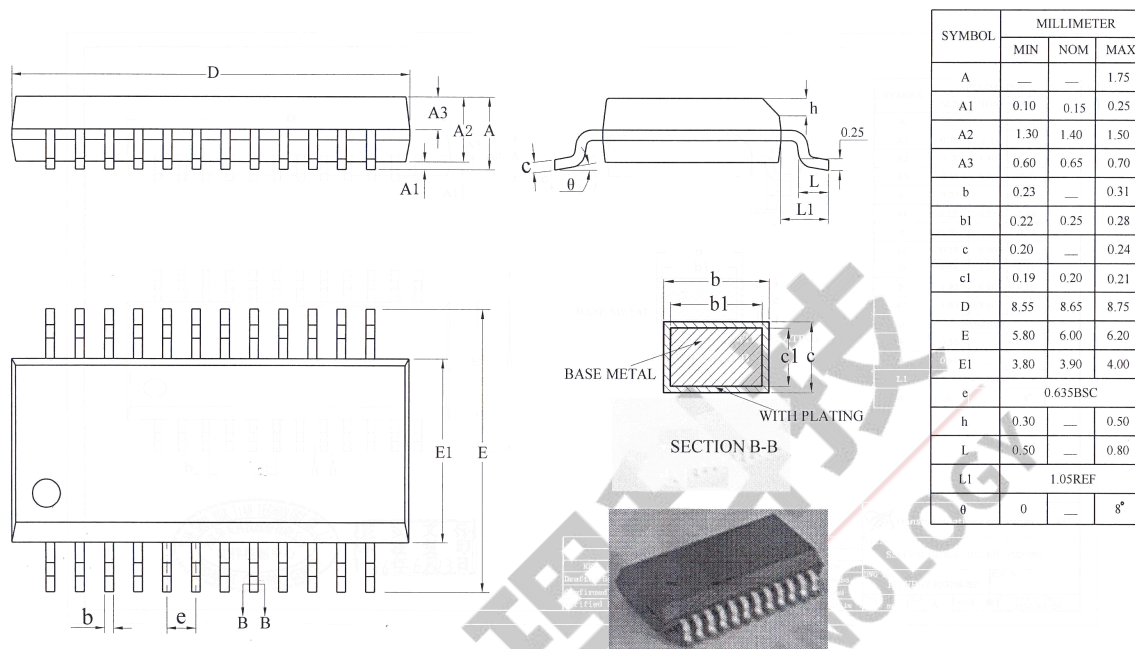
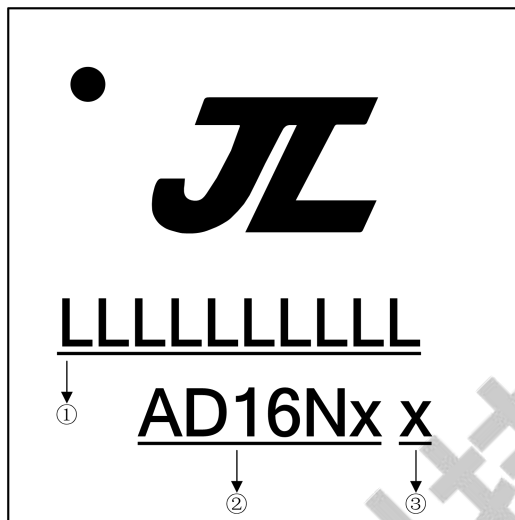


Figure 4-1 AD165D Package

5 IC Marking Information



① LLLLLLLLLL : Production Batch

② AD16Nx : Chip Model

③ Built-in flash size

0: No Flash Memory

2: 2Mbit Flash

4: 4Mbit Flash

8: 8Mbit Flash

6: 16Mbit Flash

3: 32Mbit Flash

6 Solder-Reflow Condition

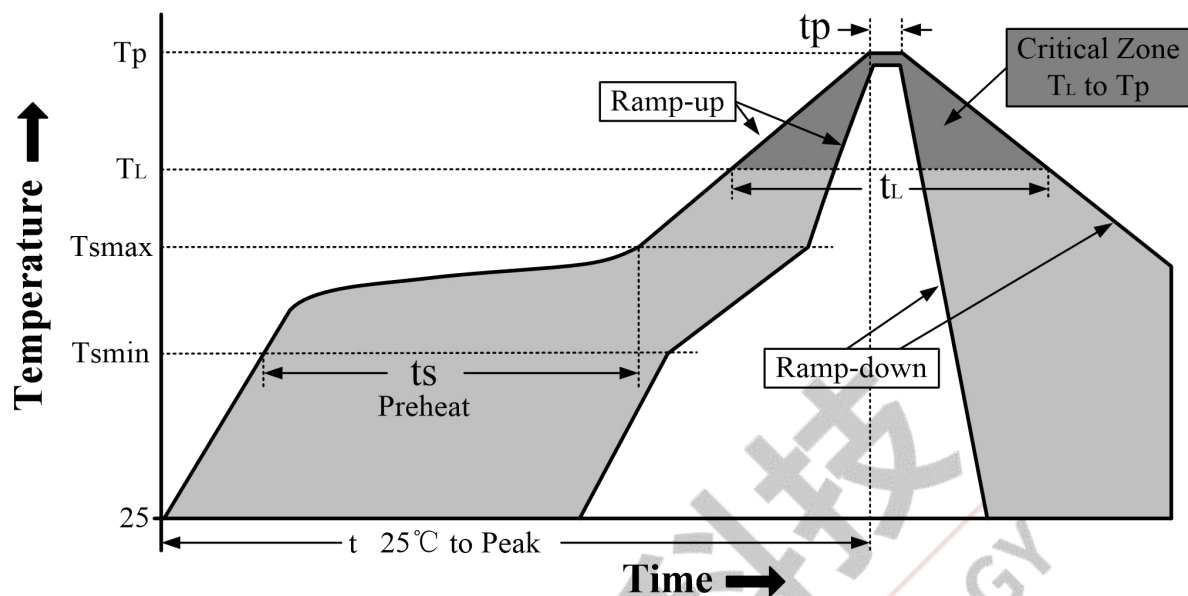


Figure 6-1 Classification Reflow Profile

Classification Profiles

Table 6-1

Profile Feature		Sn-Pb Eutectic Assembly	Pb-Free Assembly
Preheat/ Soak	Temperature Min (T_{smin})	100 °C	150 °C
	Temperature Max (T_{smax})	150 °C	200 °C
	Time (t_s) from (T_{smin} to T_{smax})	60-120 seconds	60-180 seconds
Average ramp-up rate (T_{smax} to T_p)		3 °C/second max	3 °C/second max
Liquidous temperature (T_L)		183 °C	217 °C
Time (t_L) maintained above T_L		60-150 seconds	60-150 seconds
Peak package body temperature (T_p)		See Table 6-2.	See Table 6-3.
Time within 5°C of actual Peak Temperature (t_p)		10-30 seconds	20-40 seconds
Ramp-down rate (T_p to T_L)		6 °C/second max.	6 °C/second max.
Time 25 °C to peak temperature		6 minutes max.	8 minutes max.

Note 1: All temperatures refer to topside of the package, measured on the package body surface.

Note 2: Time within 5°C of actual peak temperature (t_p) specified for the reflow profiles is a “supplier” minimum and “user” maximum.

SnPb - Classification Temperature

Table 6-2

Package Thickness	Volume mm ³ < 350	Volume mm ³ ≥ 350
<2.5 mm	240 +0/-5 °C	225 +0/-5 °C
≥ 2.5 mm	225 +0/-5 °C	225 +0/-5 °C

Pb-free - Classification Temperature **Table 6-3**

Package Thickness	Volume mm ³ < 350	Volume mm ³ 350 - 2000	Volume mm ³ > 2000
< 1.6mm	260 °C	260 °C	260 °C
1.6 mm - 2.5mm	260 °C	250 °C	245 °C
> 2.5mm	250 °C	245 °C	245 °C

7 Revision History

Date	Revision	Description
2024.04.01	V1.0	Initial Release.

